

Microscopic Traffic Flow Modelling (Part II)

Professor Zuduo Zheng
School of Civil Engineering
University of Queensland
Semester 2 2025

Agenda

- Main Car Following Models
- Car Following Models Used in Some Popular Traffic Simulation Packages

Main Car Following Models

"In the beginning there was GM."

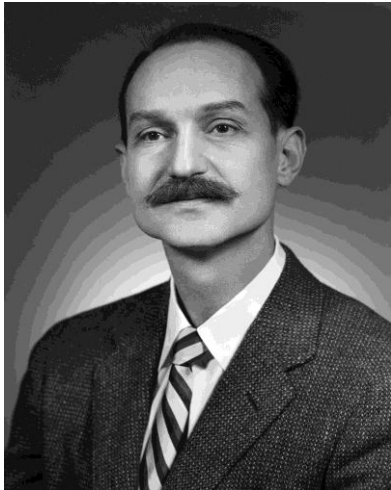
Car Following Models - GM



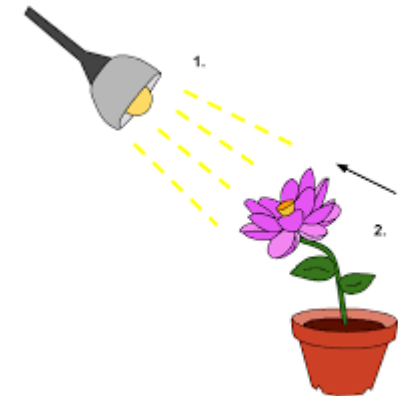
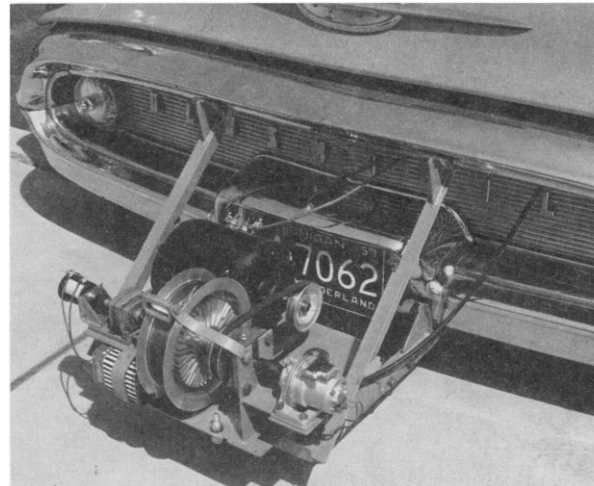
General Motors (GM) models

Back in the 1950's, General Motors sponsored a team of scientists in its Research Laboratories, from which pioneering work was done that broke the ground for traffic flow theory. At the foremost of such efforts was the family of **General Motors models (GM)**.

$$response = sensitivity \times stimulus$$

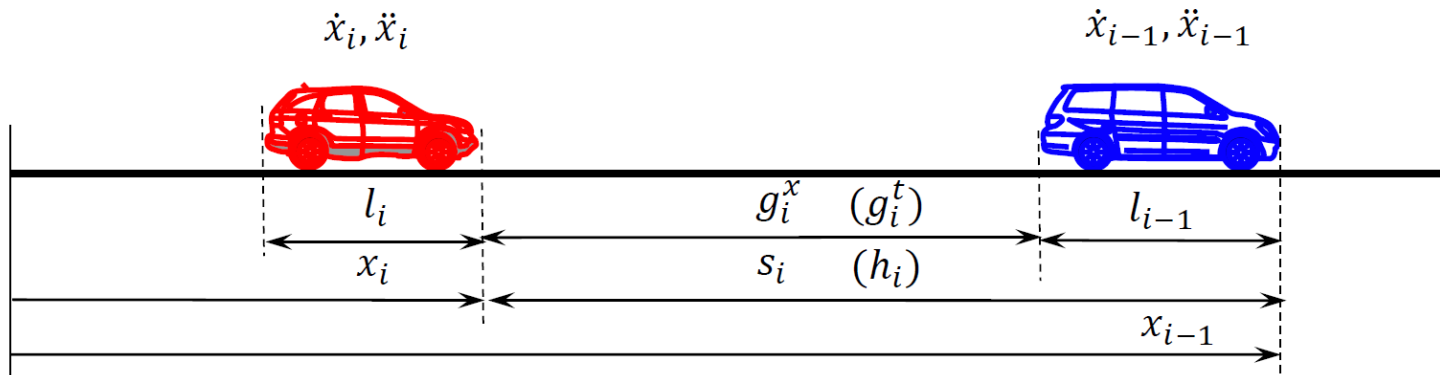


Robert Herman at GM
(Niels Bohr Library & Archives)



<https://en.wikipedia.org>

GM1



response = f (sensitivity, stimuli)

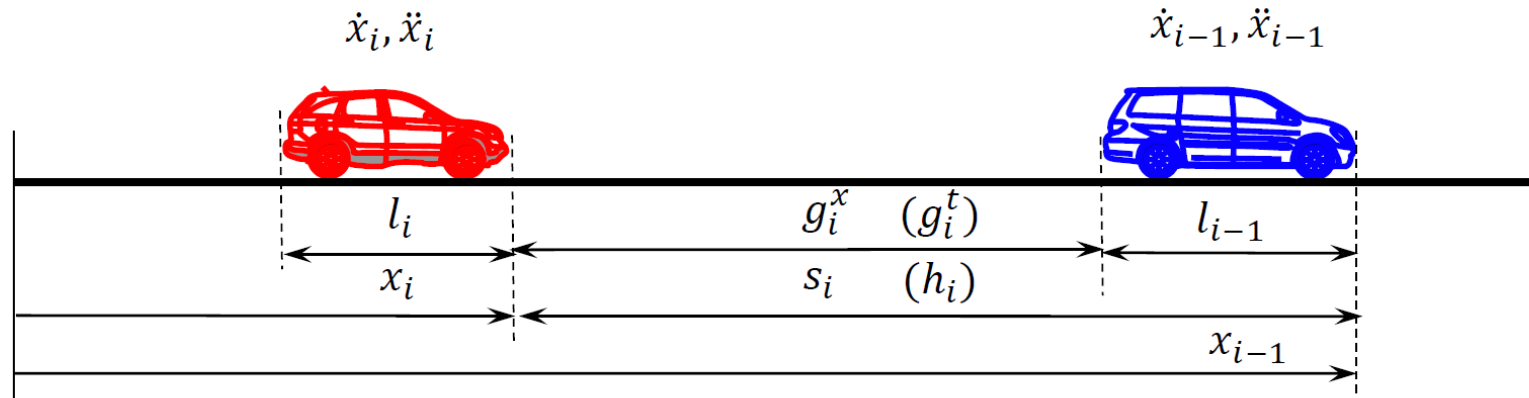
$$\ddot{x}_i(t + \tau_i) = \alpha[\dot{x}_{i-1}(t) - \dot{x}_i(t)]$$



- Scenario 1: $\dot{x}_i(t) = 120$ km/h, $\dot{x}_{i-1}(t) = 100$ km/h, and $s_i(t) = 50$ m
- Scenario 2: $\dot{x}_i(t) = 120$ km/h, $\dot{x}_{i-1}(t) = 100$ km/h, and $s_i(t) = 5000$ m

What will be the predicted accelerations in these two scenarios? Is it reasonable?

GM2



$$\ddot{x}_i(t + \tau_i) = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} [\dot{x}_{i-1}(t) - \dot{x}_i(t)]$$

Field experiments revealed that sensitivity coefficient α ranges between (0.17, 0.74). In GM2, a high sensitivity value α_1 is chosen when the two vehicles are close, while a low sensitivity value α_2 is employed when the two vehicles are far apart.

GM3

$$\ddot{x}_i(t + \tau_i) = \alpha \frac{[\dot{x}_{i-1}(t) - \dot{x}_i(t)]}{[x_{i-1}(t) - x_i(t)]}$$

- Scenario 1: In Brisbane CBD, one vehicle is following another at a spacing of 100 m with speeds $\dot{x}_i(t) = 30$ km/hr, $\dot{x}_{i-1}(t) = 10$ km/hr.
- Scenario 2: In M1 motorway, one vehicle is following another at a spacing of 100 m with speeds $\dot{x}_i(t) = 130$ km/hr, $\dot{x}_{i-1}(t) = 110$ km/hr.



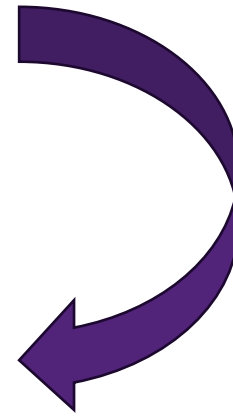
What will be the predicted accelerations in these two scenarios? Is it reasonable?

GM4

$$\ddot{x}_i(t + \tau_i) = \alpha \frac{\dot{x}_i(t + \tau_i)[\dot{x}_{i-1}(t) - \dot{x}_i(t)]}{[x_{i-1}(t) - x_i(t)]}$$



$$F = G \frac{m_1 m_2}{r^2}$$



Similarity between
GM4 and gravity
model

GM5

$$\ddot{x}_i(t + \tau_i) = \alpha \frac{[\dot{x}_i(t + \tau_i)]^m}{[x_{i-1}(t) - x_i(t)]^l} [\dot{x}_{i-1}(t) - \dot{x}_i(t)]$$

- Exponents m and l generalises GM model.
- By taking different value combinations of m and l , GM5 can be reduced to earlier models.
- GM5 serves as the “bridge” that connects microscopic and macroscopic models.

Discussion

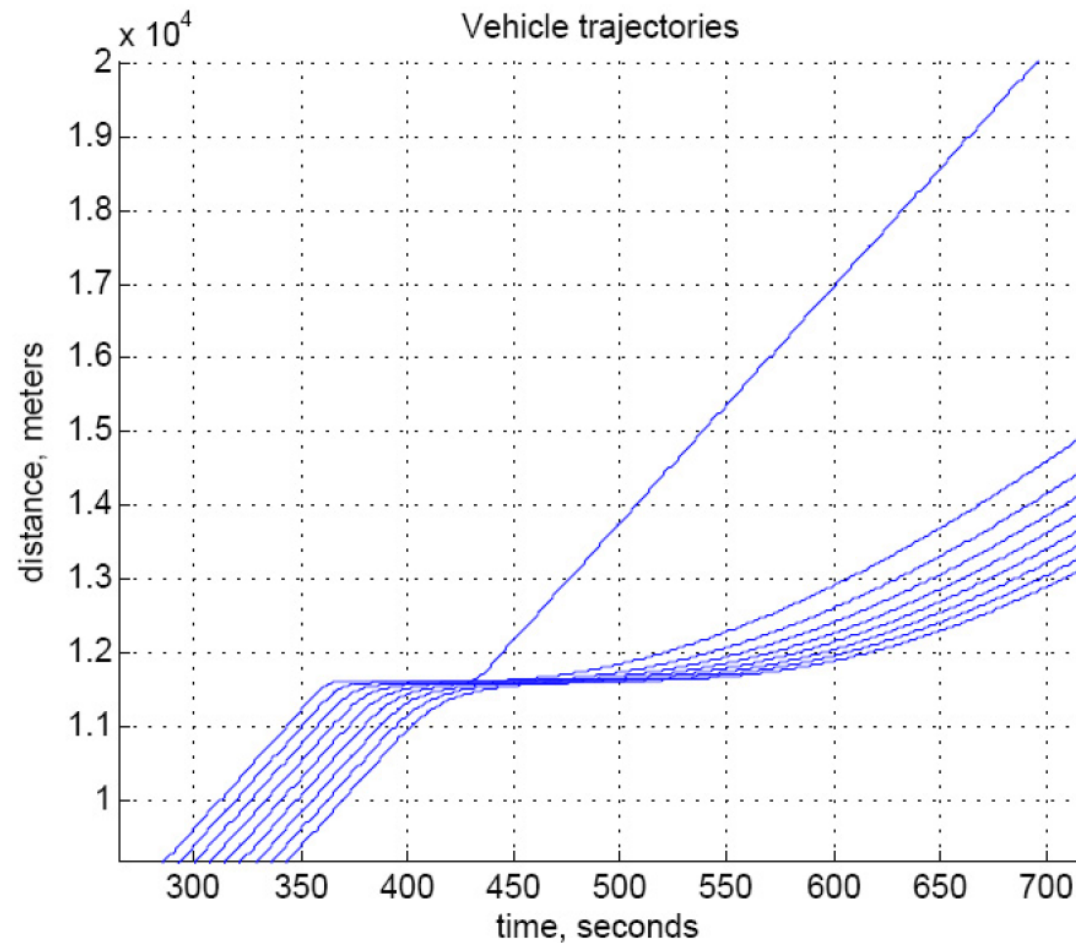
- If you use GM4 to simulate a vehicle who stops at a red traffic light and then starts moving when the traffic light turns green. What would happen?
- According to GM4, if two vehicles are travelling at the same speed, what is the implication on their spacing?



Limitations of GM Models: Slow Start

According to GM4, a vehicle at a stopped position is unable to get started. This is because the vehicle's speed ($\dot{x}_i(0) = 0$) determines its acceleration in the next step.

Therefore, the vehicle has to maintain a non-zero speed at any time in order to avoid being trapped. As such, the vehicle on red light has to slow down to an infinitesimal speed rather than a complete stop in order to avoid being trapped. When the light turns green or the leading vehicle resumes motion, the subject vehicle will take a long time to get up to speed. Figure illustrates such a scenario where a noticeable gap is resulted between the first and second vehicle.



Limitations of GM Models: An Intimate Pair

According to GM4, two vehicles can get arbitrarily close as long as they are traveling at the same speed, which is certainly not true - no one would dare to follow another with one inch apart at 120 km/h! The reason why GM4 allows such a ridiculous car-following distance is because, regardless of how close the two vehicles are, the model predicts no response as long as the two vehicles are moving at the same speed.



Extensions of GM Models

- **Memory functions:** Assuming that a driver reacts to the relative speed of the preceding vehicle over a period of time, rather than in an instant, a memory function was introduced into the linear GM model to store the information of relative speed during CF. Although the model removes unrealistic peaks in acceleration profile, the implementation of the model in traffic simulation is considerably more complex due to the need of maintaining an array of past conditions for each vehicle.
- **Acceleration and deceleration asymmetry:** 1) most passenger cars have a greater deceleration than acceleration capacity. And 2) in congested traffic, drivers are more sensitive to deceleration than to acceleration.
- **Multiple-vehicle interaction:** The models discussed above are based on the assumption that each driver reacts in some specific manner to some stimuli from the preceding vehicle. In the real world, however, drivers most likely adjust their behaviors according to their observations of more than one vehicle ahead.

CF Models Used in Some Popular Traffic Simulation Packages

Car Following Models – Gipps' model



Transportation Research Part B: Methodological

Volume 15, Issue 2, April 1981, Pages 105-111



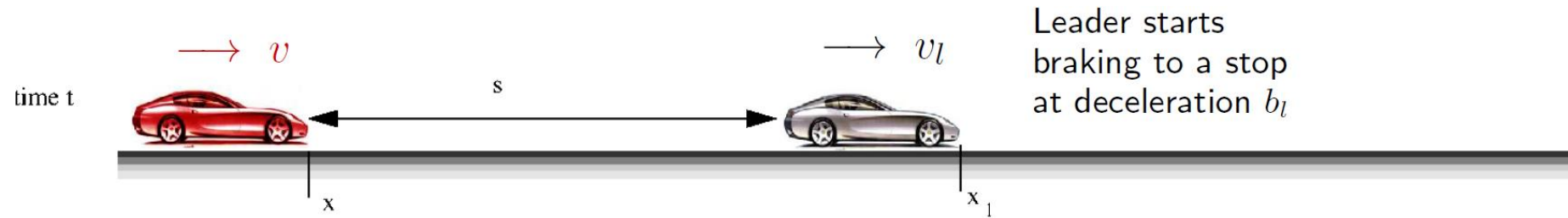
A behavioural car-following model for computer simulation

P.G. Gipps

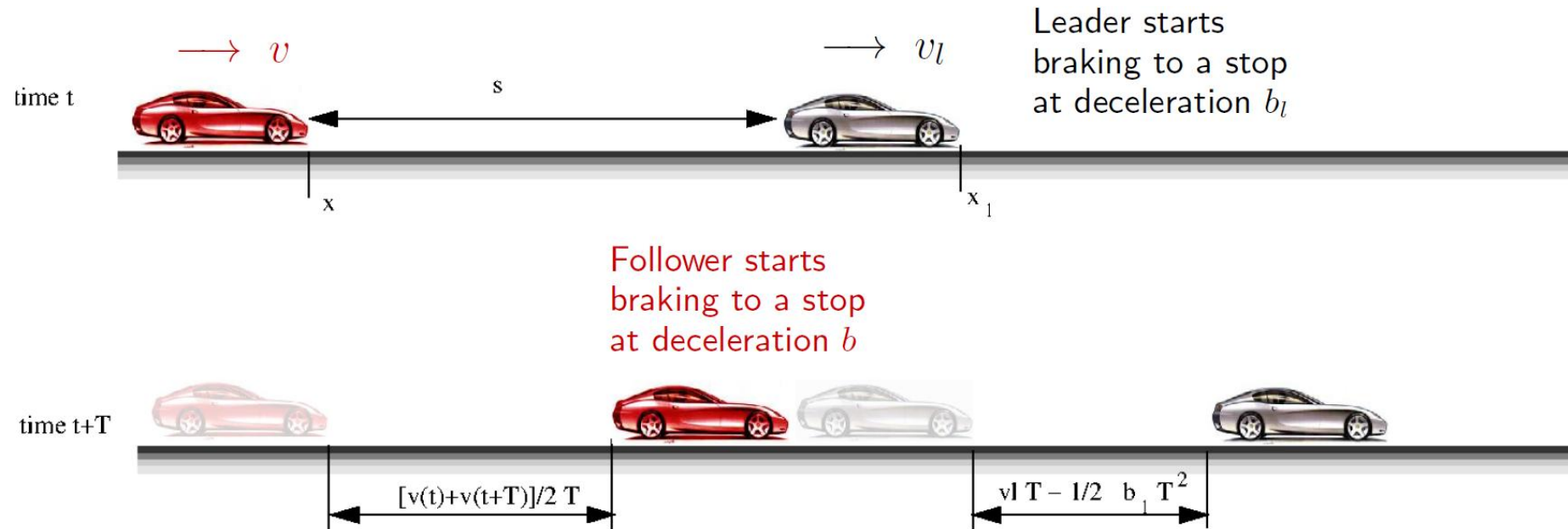
Gipps' model

- Gipps' model is the most popular safety distance model and developed by Gipps in 1981.
- Safety distance models differ from GM models by hypothesizing that the **driver reacts to spacing relative to the preceding vehicle, rather than to the relative speed**. This idea was first proposed by Kometani and Sasaki (1959).
- Gipps' model assumes that the speed is selected by the driver in a way to ensure that **the vehicle can be safely stopped in case the preceding vehicle should suddenly brake**.
- Gipps' model includes two modes of driving: free-flow and CF. The driver chooses the smaller one from the speeds obtained from the free-flow and CF modes.

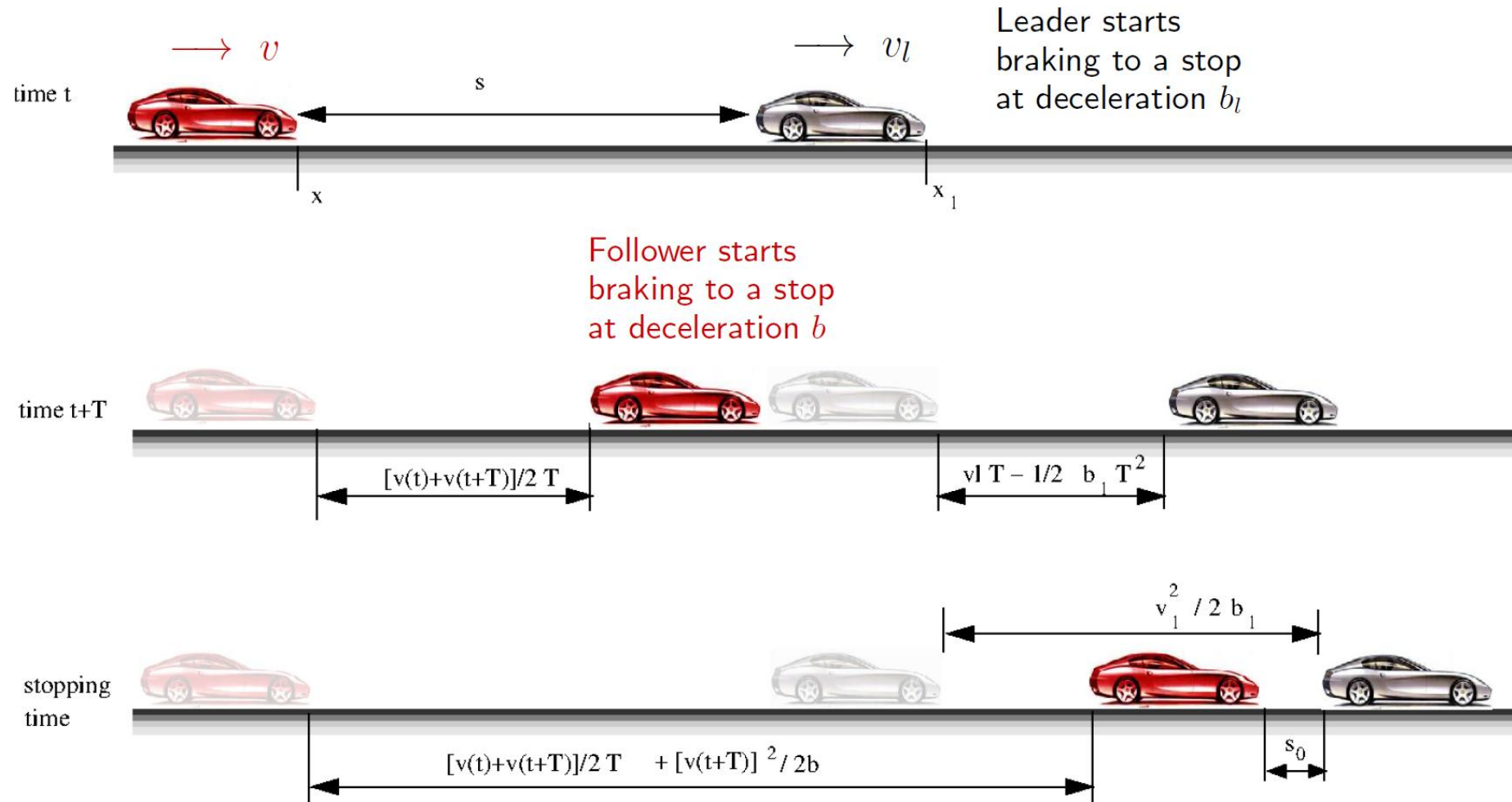
Derivation of Gipps' CF Model: Overview



Derivation of Gipps' CF Model: Overview



Derivation of Gipps' CF Model: Overview



Gipps' CF model

□ Modeling approach: considered limitations/constraints of vehicles.

➤ In **free-flow case**:

- It will not exceed its driver's desired speed.
- Its free acceleration should first increase with speed as engine torque increases then decrease to zero as the desired speed is reached.

$$v_n(t + \tau) \leq v_n(t) + 2.5a_n\tau(1 - v_n(t)/V_n)(0.025 + v_n(t)/V_n)^{1/2}$$

Variables

- a_n - max acceleration that vehicle n wishes to take
- V_n - desired speed of vehicle n
- τ is the reaction time of vehicle n .

Gipps' CF model

- In **braking case**: To assure that if the leader brakes, the vehicle will not crash. This means that we should guarantee the **worst, but still safe, condition**; i.e., the subject vehicle is about to crash when it comes to stop.

- This requires the following inequality:

$$x_{n-1}^* = x_{n-1}(t) - v_{n-1}(t)^2 / 2b_{n-1}$$

$$x_n^* = x_n(t) + [v_n(t) + v_n(t + \tau)] * \frac{\tau}{2} + v_n(t + \tau)\theta + \frac{v_n(t + \tau)^2}{-2b_n}$$

$$x_{n-1}^* - s_{n-1} \geq x_n^*$$

- τ : subject vehicle reaction time. θ is the delay in addition to the regular reaction time τ . This represents the safety margin.
- b_n : subject vehicle deceleration;
- b_{n-1} : leader deceleration; the subject vehicle has to estimate b_{n-1} . We use $\hat{b}$ to replace that.
- Both b_n and b_{n-1} are negative if vehicles are decelerating.
- s_{n-1} : the jam spacing, usually s_{n-1} equals to vehicle length and some extra spacing for buffer.
- If $\theta = \tau/2$, we can solve the inequality above to obtain speed $v_n(t + \tau)$:

Gipps' CF model

➤ In **braking case**:

$$x_{n-1}^* = x_{n-1}(t) - v_{n-1}(t)^2 / 2b_{n-1}$$

$$x_n^* = x_n(t) + \underbrace{[v_n(t) + v_n(t + \tau)] * \frac{\tau}{2}}_{\text{term1}} + \underbrace{v_n(t + \tau)\theta}_{\text{term2}} + \underbrace{\frac{v_n(t + \tau)^2}{-2b_n}}_{\text{term3}}$$

$$x_{n-1}^* - s_{n-1} \geq x_n^*$$

- **term1**: during the reaction time, the subject vehicle will continue its previous kinematic dynamics (e.g., acceleration or deceleration). Therefore, term 1 captures the traveled distance during from $[t, t + \tau]$.
- **term2**: at $(t + \tau)$, the subject vehicle has decided its next action (deceleration/acceleration), but it may have some delay in executing that, denoted as θ . During $[t + \tau, t + \tau + \theta]$, the subject vehicle cruises at its speed at $v_n(t + \tau)$. Term 2 is the traveled distance during this process.
- **term 3**: from time $(t + \tau + \theta)$, subject vehicle starts to decelerate at b_n until it stops. Note that there is a negative sign because it's decelerating. b_n is negative. Term 3 is the travel distance during this process.

Gipps' CF model

➤ In **braking case**:

$$x_{n-1}^* = x_{n-1}(t) - v_{n-1}(t)^2 / 2b_{n-1}$$

$$x_n^* = x_n(t) + \underbrace{[v_n(t) + v_n(t + \tau)] * \frac{\tau}{2}}_{\text{term1}} + \underbrace{v_n(t + \tau)\theta}_{\text{term2}} + \underbrace{\frac{v_n(t + \tau)^2}{-2b_n}}_{\text{term3}}$$

$$x_{n-1}^* - s_{n-1} \geq x_n^*$$

To assure that if the leader brakes, the vehicle will not crash.

- **term1**: during the reaction time, the subject vehicle will continue its previous kinematic dynamics (e.g., acceleration or deceleration). Therefore, term 1 captures the traveled distance during from $[t, t + \tau]$.
- **term2**: at $(t + \tau)$, the subject vehicle has decided its next action (deceleration/acceleration), but it may have some delay in executing that, denoted as θ . During $[t + \tau, t + \tau + \theta]$, the subject vehicle cruises at its speed at $v_n(t + \tau)$. Term 2 is the traveled distance during this process.
- **term 3**: from time $(t + \tau + \theta)$, subject vehicle starts to decelerate at b_n until it stops. Note that there is a negative sign because it's decelerating. b_n is negative. Term 3 is the travel distance during this process.

Gipps' CF model

- Solving the inequality gives us:

$$v_n(t + \tau) \leq b_n \left(\frac{\tau}{2} + \theta \right) + \sqrt{b_n^2 \left(\frac{\tau}{2} + \theta \right)^2 - b_n \left[2s - v_n(t)\tau - \frac{v_{n-1}(t)^2}{\hat{b}} \right]}$$

where $s = x_{n-1}(t) - x_n(t) - s_{n-1}$, is the net-spacing (i.e., excluding jam spacing)

- If $\theta = \tau/2$, speed $v_n(t + \tau)$ becomes:

$$v_n(t + \tau) \leq b_n \tau + \sqrt{b_n^2 \tau^2 - b_n \left[2s - v_n(t)\tau - \frac{v_{n-1}(t)^2}{\hat{b}} \right]}$$

Gipps' CF model is the base model in Aimsun.



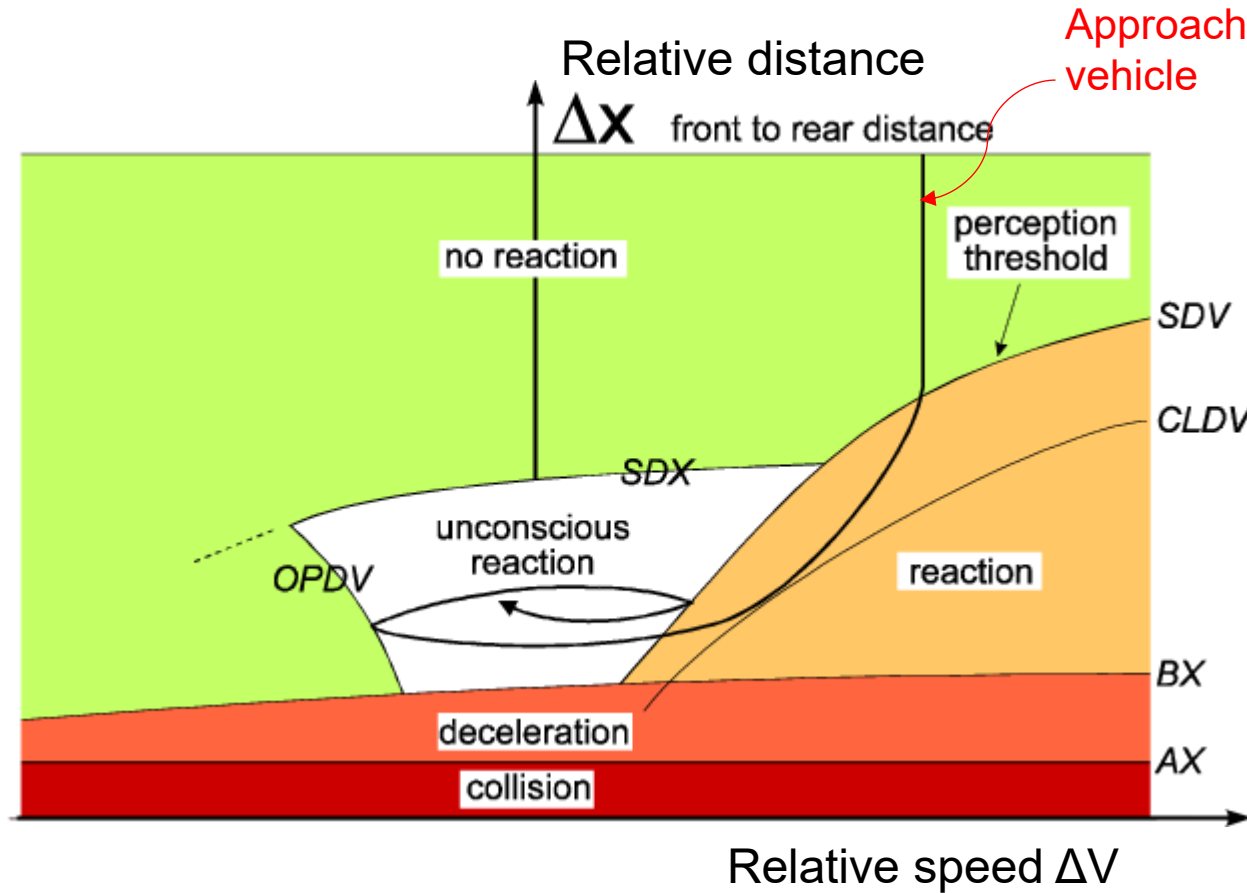
Discussion

**Which type does Gipps CF model belong to?
Continuous time or discrete time?**



Wiedemann's CF model/'psycho-physical' models

- Engineering CF models unrealistically assume that drivers can perceive and react even to small changes in the driving environment (for example, to slight change in speed difference or spacing).
- To overcome this problem, Wiedemann (1974) introduces the term 'perceptual threshold' to define the minimum value of the stimulus a driver can perceive and will react to.
- The models based on perceptual threshold are also known as 'psycho-physical' models. The threshold is expressed as a function of speed difference and spacing between the preceding and subject vehicles, and is different for acceleration and deceleration decisions. It increases driver alertness when spacing is small, and provides more freedom when it is large.



Wiedemann's CF model (Source: Wiedemann, 1974)

AX: The desired spacing between the front sides of two successive vehicles in a standing queue

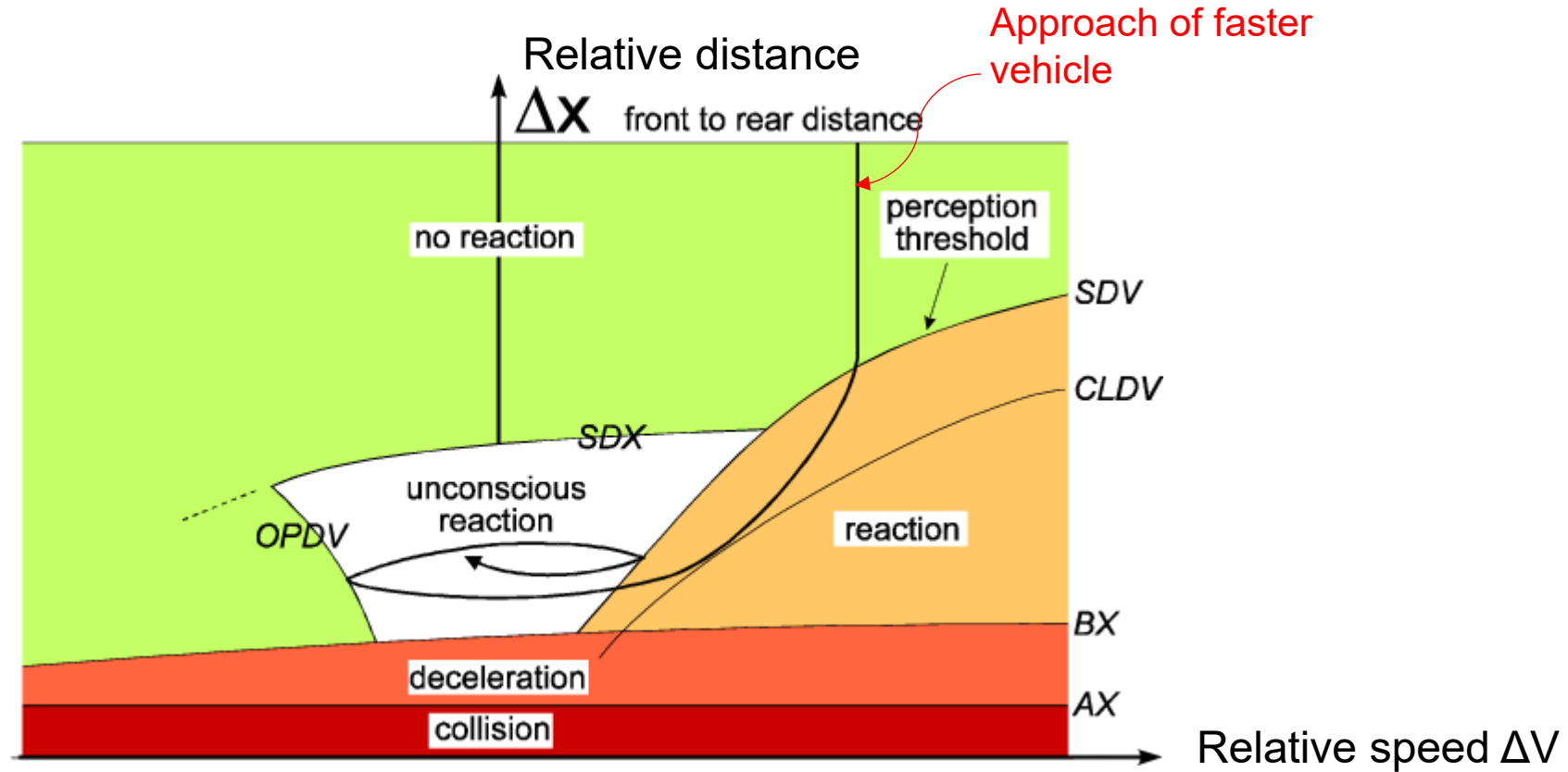
BX: The desired minimum following distance, which is a function of AX, the safety distance, and speed

SDV: The action point where a driver consciously observes that he/she is approaching a slower leading vehicle; SDV increases with increasing speed difference

CLDV: Closing delta velocity (CLDV) is an additional threshold that accounts for additional deceleration by the application of brakes

OPDV: The action point where a driver notices that he/she is slower than the leading vehicle and starts to accelerate again

SDX: A perception threshold to model the maximum following distance, which is approximately 1.5–2.5 times BX



The dark line in the figure shows the decision path of an approaching vehicle. A vehicle travelling faster than the leader will get close to it until the deceleration perceptual threshold (SDV) is crossed. The driver will then decelerate to match the leader's speed. However, as a human being, the driver is unable to accurately replicate the leader's speed, and spacing will increase until the acceleration perceptual threshold (OPDV) is reached. The driver will again accelerate to match the leader's speed and the process continues, as shown in the unconscious reaction zone.

A modified version of the original Wiedemann model has been used in the commercial microsimulation software **VISSIM**.



Intelligent driver model (IDM)

One of the most popular models using desired measures is the intelligent driver model (IDM) proposed by Treiber et al. (2000). This model considers both the desired speed and the desired space headway, as defined in

$$\dot{v} = a \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right] \quad \text{IDM.}$$

where $a_{\max}^{(n)}$ is the maximum acceleration of the subject vehicle n , v_0 is the desired speed, S is spacing between two vehicles measured from the front edge of the subject vehicle to the rear end of the preceding vehicle, s^* is the desired spacing, and δ is a parameter.

Intelligent driver model (IDM)

- When preceding vehicle is far away, the third term in this equation becomes negligible and the model performs as a free flow model where the desired speed of the driver governs the acceleration.
- Use of one equation ensures a smooth transition between free-flow and car-following situations.
- The desired space headway (or following distance) in IDM is dependent on several factors and can be calculated using

$$s^*(v, \Delta v) = s_0 + \max \left(0, vT + \frac{v\Delta v}{2\sqrt{ab}} \right)$$

Discussion

- a) Compared with GM models and Gipps' CF model, which parameter commonly used in many other CF models is not used in IDM?
- b) If $\delta \rightarrow \infty$, which model (GM 1~4, Gipps' CF model, or Wiedemann) will IDM become similar to? Why?



Interpreting IDM

We can easily interpret the model parameters by considering the following three standard situations:

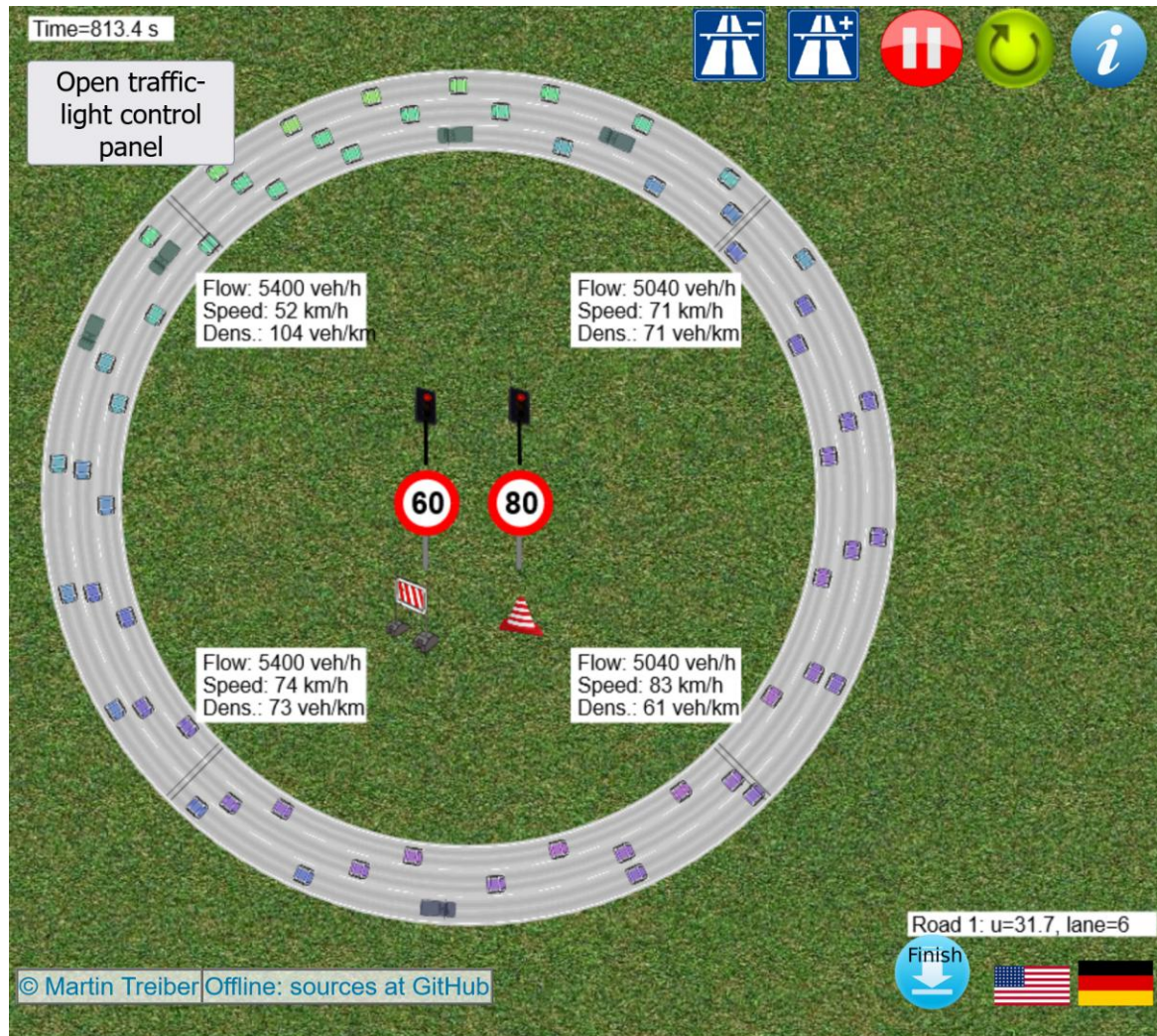
- When *accelerating on a free road from a standstill*, the vehicle starts with the maximum acceleration a . The acceleration decreases with increasing speed and goes to zero as the speed approaches the desired speed v_0 . The exponent δ controls this reduction: The greater its value, the later the reduction of the acceleration when approaching the desired speed.
- When *following a leading vehicle*, the distance gap is approximatively given by the *safety distance* $s_0 + vT$.
- When *approaching slower or stopped vehicles*, the deceleration usually does not exceed the comfortable deceleration b .
- The acceleration function is smooth during transitions between these situations.

IDM is implemented in SUMO



SUMO (Simulation of Urban MObility), an open source, highly portable, microscopic and continuous multi-modal traffic simulation package designed to handle large networks.

Online Traffic Simulating Using IDM



Traffic Flow and General

Density/lane 26 veh/km
Truck Perc 10 %
Timelapse 6 times

Car-Following Behavior

Max Accel a 1.5 m/s²
Max Speed v_0 108 km/h
Time Gap T 1.4 s
Min Gap s_0 2 m
Comf Decel b 3 m/s²

Lane-Changing Behavior

Politeness 0.1 m/s²
LC Threshold 0.4 m/s²
Right Bias Cars 0.05 m/s²
Right Bias Trucks 0.2 m/s²

Further readings:

- Chapter 10, Traffic flow dynamics: Data, models and simulation by Martin Treiber and Arne Kesting.
- van Wageningen-Kessels, F., Van Lint, H., Vuik, K., & Hoogendoorn, S. (2015). Genealogy of traffic flow models. *EURO Journal on Transportation and Logistics*, 4(4), 445-473.
- Saifuzzaman, M., Zheng, Z., 2014. Incorporating human-factors in car-following models: A review of recent developments and research needs. *Transportation Research Part C* 48, 379 - 403.



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

Thank you

Questions?